A closed-form approximation to the biomarker-treatment interaction estimator in aggregated N-of-1 trials, and what it implies for design efficiency under a mandatory open-label lead-in

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Contents

Short communication.

1 Introduction

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Every design question this project's companion manuscripts have addressed — how many randomization paths, what sample size, how long to wait before a discontinuation-phase assessment, whether a brief crossover tail is worth its extra visits — has so far been answered by Monte Carlo simulation[1–5]. That is appropriate for the compendium's headline estimand, which combines a Gompertz response trajectory, a dual-channel biomarker-moderation architecture, exponential or Weibull carryover, and a continuous-time AR(1) mixed model into a system with no obvious closed form. But it leaves a practical gap: a trialist choosing among two or three concrete design options today cannot get even an order-of-magnitude power estimate without running a fresh simulation, in contrast to the treatment main effect, for which Senn's and Zucker's classical variance formulas[6,7] have long supplied exactly this kind of design intuition, and in contrast to the dichotomized-biomarker case, for which the companion test-procedure manuscript derives a closed-form repeated-measures ANOVA power formula[8]. Neither covers the compendium's actual estimand: a continuous biomarker, a continuous exposure predictor D_{bc}, and a mixed model fit by `nlme::lme` with `corCAR1` residual correlation.

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This short communication was motivated by a concrete instance of the gap: a trial whose first phase must be open-label, for recruitment reasons that are not themselves a design choice. Given that fixed constraint, what should the analyst prioritize among the remaining design parameters to make the biomarker-treatment interaction test as efficient as possible? We first give a qualitative answer, drawn directly from the compendium's existing simulation results. We then ask whether a genuine closed-form approximation to the interaction estimator's sampling variance and power can be derived, and, if so, how far it can be trusted. We derive such an approximation from two components: a within-subject two-group contrast reduction of the analysis model, and the exact variance of a mean of AR(1)-correlated, unequally spaced observations. We validate both components against Monte Carlo output already on disk, produced for a companion manuscript, rather than commissioning a new simulation study — the appropriate scope for a short communication. We report the validation honestly: the variance and power approximation performs well: the attenuation-factor component captures the correct qualitative ranking of designs by carryover vulnerability but not its magnitude. We close with quantitative, closed-form-grounded design guidance for the open-label-first scenario.

2 What the compendium already establishes

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Before deriving anything new, it is worth stating precisely what the existing simulation evidence already supports, since a closed form is only useful to the extent that it reproduces conclusions already established with more care. Four results from the compendium bear directly on the open-label-first design question.

First, design topology after the open-label phase is the largest lever available. At matched total sample size $N = 70$, the Hybrid design (open-label, then blinded discontinuation, then a brief crossover; four randomization paths) dominates open-label-plus- blinded-discontinuation alone (OL+BDC; two paths) in absolute power at every carryover half-life tested, 0.842 versus 0.751 at $t_{1/2} = 0$ and 0.784 versus 0.730 at $t_{1/2} = 1.0$ week[1]. Adding the crossover tail is worth its extra visits.

Second, matching total sample size N across designs being compared is not optional bookkeeping: it changed the compendium's own conclusions twice. An unmatched comparison inflated Hybrid's apparent power ceiling in the architecture manuscript and manufactured an apparent dropout-robustness advantage for Hybrid in the informative-dropout manuscript that a matched re-run showed was a ceiling effect[1,4].

Third, the carryover decay's functional *shape* (exponential versus Weibull, across shape parameters $k \in \{0.25, 0.5, 2, 4\}$) barely moves the compendium's substantive conclusions; the *analysis specification* — whether the exposure predictor is a binary indicator, an exposure-weighted continuous decay, or an AIC-selected-half-life variant — matters considerably more[2].

Fourth, misspecifying the residual correlation structure (fitting an exchangeable or naive model when the truth is AR(1)) is a larger anti-conservatism risk than misspecifying carryover decay shape[8].

These four results already answer "what to prioritize" at a qualitative level. What they do not supply is a way to move from a verbal ranking to a number: how much does adding the crossover tail buy, quantitatively, as a function of the design's own visit schedule, without running a new simulation for every candidate design? That is the question the rest of this communication addresses.

3 A closed-form approximation

3.1 Setup

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We work under the mean-moderation architecture[1], in which the biomarker B_i scales the treatment effect directly in the outcome's conditional mean rather than through a covariance channel, and we approximate the analysis specification E1 (binary on-drug indicator D_{it}, carryover unmodeled at the analysis stage;[2]) by a simple two-group within-subject contrast. For participant i with k_{on} on-drug and

k_{off} off-drug assessment timepoints at fixed calendar times, define

$$\Delta_i = \overline{Y}_i^{on} - \overline{Y}_i^{off}, \tag{1}$$

the mean on-drug outcome minus the mean off-drug outcome. Because the participant random intercept u_i is common to both means, it cancels in Δ_i exactly (no random slope is present in the compendium's baseline model), leaving

$$\Delta_i = \beta_1(1+\beta_{bm}B_i)\cdot \bar{D}_{bc,i}^{on} - \beta_1(1+\beta_{bm}B_i)\cdot \bar{D}_{bc,i}^{off} + \eta_i, \tag{2}$$

where $\bar{D}_{bc,i}^{on} = 1$ by construction and $\bar{D}_{bc,i}^{off} = \overline{D_{bc_i}}^{off}$ is the mean of the continuous exposure-decayed indicator across the off-drug assessment times, which is less than one whenever residual carryover has not fully decayed by the time of assessment, and η_i is the difference of the two residual means. A second-stage ordinary least squares regression of Δ_i on the standardized biomarker b_i then recovers the interaction coefficient as its slope,

$$\Delta_i = \gamma_0 + \beta_{bm:D}\, b_i + \eta_i, \qquad \beta_{bm:D} = \beta_1\, \beta_{bm}\cdot A_i, \tag{3}$$

where $A_i = 1 - \overline{D_{bc_i}}^{off}$ is a per-participant **carryover attenuation factor**. This construction is exactly the two-stage paired-difference specification E9 of the companion carryover manuscript[2], specialized to a strict on/off split rather than E9's general on/off means, and is offered here as an approximation to E1 (which is fit by the full mixed model on all timepoints jointly, not by this two-group reduction) rather than as an exact restatement of it.

3.2 Sampling variance under AR(1) correlation

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The variance of the mean of k observations drawn from a stationary process with autocovariance $\gamma(h) = \sigma^2\rho^{|h|}$ at arbitrary, possibly unequally spaced times $t_1, \dots, t_k$ has the exact closed form

$$V(\mathbf{t}, \sigma^2, \rho) = \frac{\sigma^2}{k^2}\sum_{j=1}^{k}\sum_{l=1}^{k}\rho^{|t_j - t_l|}. \tag{4}$$

Applying Equation (4) separately to a participant's on-drug and off-drug visit times and summing (treating the two period means as approximately independent, an approximation we return to in the Discussion) gives the variance of the within-subject contrast,

$$\mathrm{Var}(\eta_i) \approx V(\mathbf{t}^{on}, \sigma^2, \rho) + V(\mathbf{t}^{off}, \sigma^2, \rho) \;\equiv\; v(\text{design}, \rho), \tag{5}$$

a quantity computable directly from a design's visit schedule and an assumed

AR(1) parameter ρ, with no simulation required. Pooling across N participants (assuming a standardized biomarker, so $\mathrm{Var}(b_i) = 1$) and, for a multi-path design, averaging $v(\text{design}, \rho)$ across the design's randomization paths, the closed-form approximation to the interaction estimator's sampling variance is

$$\widehat{\mathrm{Var}}(\hat{\beta}_{bm:D}) \approx \frac{\sigma^2\, v_1(\text{design}, \rho)}{N}, \tag{6}$$

where v_1 is $v(\text{design}, \rho)$ evaluated at $\sigma^2 = 1$, so that σ^2 enters as a single free scale parameter separating design-and-correlation structure (v_1, closed-form) from residual noise magnitude (σ^2, an empirical quantity like any other nuisance variance component).

3.3 Predicted power

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Given Equations (3) and (6), a Wald-normal approximation to power at level α follows immediately, in the same spirit as the general four-channel decomposition of the companion calibration manuscript[8, its $\kappa = 1, \nu = \infty$ special case]:

$$z = \frac{|\beta_{bm:D} \cdot A(t_{1/2})|}{\sqrt{\sigma^2\, v_1(\text{design}, \rho)/N}}, \qquad \hat{\pi} \approx \Phi(z - z_{1-\alpha/2}) + \Phi(-z - z_{1-\alpha/2}), \tag{7}$$

with $A(t_{1/2})$ the design-averaged attenuation factor of Equation (3) and $\beta_{bm:D}$ the interaction coefficient's magnitude in the absence of carryover. Every quantity on the right-hand side except σ^2, ρ, and $\beta_{bm:D}$ itself is computable directly from the candidate design's visit schedule and a candidate carryover half-life, with no simulation.

4 Validation against existing Monte Carlo output

4.1 Data and procedure

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We validate Equations (6) and (7) against Monte Carlo output already produced for the dual-channel-architecture companion manuscript[5], rather than running a new simulation. That manuscript's driver (`analysis/scripts/dgp-combined/01-run-combined-factor` generates data under a combined architecture with independent mean-channel and covariance-channel weights $c_{bm,A}$ and $c_{bm,B}$; its ($c_{bm,A} = 0.45$, $c_{bm,B} = 0$) boundary is, by construction, exactly the pure mean-moderation architecture this communication's derivation assumes. Its stored summary (`analysis/data/dgp-combined/01-combined-summary.rds`) already includes matched total sample sizes $N \in \{35, 70\}$, three designs (crossover, CO; open-label-plus-blinded-discontinuation, OL+BDC; Hybrid), three carryover half-lives $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks, and analysis specification E1, at 1,000 replicates per cell. We compute $v_1(\text{design}, \rho{=}0.7)$ and $A(\text{design}, t_{1/2})$ directly from the same visit schedules the driver itself uses[2](sec2) for the design definitions, reproduced with driver-matching timepoints in `analysis/scripts/nof1-design-efficiency/01-clos` fit the single free scale parameter σ^2 by ordinary least squares of $\widehat{\mathrm{sd}}(\hat\beta)^2 \cdot N$ on v_1 across all 18 design-by-half-life-by-N cells, and then compare the resulting closed-form power predictions to the cells' simulated power. $\rho = 0.7$ is the AR(1) parameter the driver itself uses by default and is not separately estimated.

4.2 The variance-scaling component performs well

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With a single fitted scale parameter ($\hat\sigma^2 = 0.667$), the closed form explains $R^2 = 0.948$ of the variation in $\widehat{\mathrm{sd}}(\hat\beta)^2 \cdot N$ across the 18 cells (Table 1). Two structural predictions of Equation (6) are confirmed directly: the sampling variance scales as $1/N$ almost exactly (the CO design's empirical standard deviation ratio between $N = 35$ and $N = 70$ is 1.439, against a predicted $\sqrt{70/35} = 1.414$), and the design ranking by v_1 (OL+BDC highest, Hybrid intermediate, CO lowest, at fixed ρ) is reproduced in direction, though not exactly in magnitude, by the empirical standard deviations. A systematic residual pattern is visible and is reported rather than concealed: the closed form under-predicts CO's empirical standard deviation by roughly 15-20% and over-predicts Hybrid's and OL+BDC's by roughly 5-10%. The most likely source is Equation (5)'s independence approximation between a participant's on-drug and off-drug period means, which is exact only when no AR(1) correlation crosses the on/off boundary; CO's two long, contiguous four-visit blocks per path plausibly violate that approximation more than the other designs' more fragmented schedules.

Table 1: Closed-form sampling-standard-deviation predictions against Monte Carlo output, at $\hat{\sigma}^2 = 0.667$ (fitted once, across all 18 cells). $\mathrm{sd}(\hat{\beta})$ is the empirical standard deviation of the fitted `bm:Dbc` coefficient across 1,000 replicates; $\widehat{\mathrm{sd}}(\hat{\beta})$ is the closed-form prediction of Equation 6.

Design	N	$t_{1/2}$ (wk)	$\mathrm{sd}(\hat{\beta})$	$\widehat{\mathrm{sd}}(\hat{\beta})$	Ratio
CO	35	0.0	0.159	0.132	1.21
CO	70	0.0	0.110	0.093	1.18
Hybrid	35	0.0	0.123	0.134	0.92
Hybrid	70	0.0	0.084	0.095	0.89
OL+BDC	35	0.0	0.138	0.147	0.93
OL+BDC	70	0.0	0.096	0.104	0.92
CO	35	1.0	0.153	0.132	1.16
CO	70	1.0	0.108	0.093	1.16
Hybrid	35	1.0	0.120	0.134	0.89
Hybrid	70	1.0	0.087	0.095	0.92
OL+BDC	35	1.0	0.139	0.147	0.94
OL+BDC	70	1.0	0.096	0.104	0.92

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The $t_{1/2} = 0.5$ rows, omitted from Table 1 for space, follow the same pattern and are included in the full 18-row output of `analysis/data/nof1-design-efficiency/closed-form-valida`

4.3 Power prediction

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Using the fitted $\hat{\sigma}^2$ and, for each cell, that cell's own empirical $\bar{\hat{\beta}}$ as the effective true effect in Equation (7) — a choice discussed in §6 — the closed form predicts simulated power with a mean absolute error of 0.049, a root-mean-square error of 0.058, and a correlation of 0.935 across the 18 cells (Table 2 gives the $t_{1/2} = 0$ and $t_{1/2} = 1.0$ rows in full). For a first-pass closed form validated against a single architecture at one ρ value, this is a substantively useful level of agreement: a trialist screening candidate designs would reach the same qualitative decision from the closed form as from the full Monte Carlo simulation in the great majority of cells inspected here.

Table 2: Closed-form power predictions against simulated power (1,000 replicates per cell, $\alpha = 0.05$ two-sided).

Design	N	$t_{1/2}$ (wk)	Power (simulated)	Power (closed form)	Error
CO	35	0.0	0.352	0.421	+0.069
CO	70	0.0	0.595	0.690	+0.095
Hybrid	35	0.0	0.452	0.426	−0.026
Hybrid	70	0.0	0.738	0.703	−0.035
OL+BDC	35	0.0	0.335	0.367	+0.032
OL+BDC	70	0.0	0.612	0.618	+0.006
CO	35	1.0	0.323	0.409	+0.086
CO	70	1.0	0.611	0.716	+0.105
Hybrid	35	1.0	0.456	0.418	−0.038
Hybrid	70	1.0	0.729	0.688	−0.041
OL+BDC	35	1.0	0.327	0.357	+0.030
OL+BDC	70	1.0	0.577	0.596	+0.019

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The closed form correctly identifies Hybrid as the best-powered design at both sample sizes and both half-lives shown, matching the compendium's own headline finding[1]. It does not correctly order CO against OL+BDC: the closed form ranks CO above OL+BDC at $N = 70$ (predicted power 0.690 versus 0.618 at $t_{1/2} = 0$), while the simulation ranks them the other way (0.595 versus 0.612). This is a genuine limitation, not a rounding artefact, and is reported as such rather than smoothed over: the closed form is reliable for identifying the best design among candidates that differ substantially in v_1, and unreliable for close calls between designs whose v_1 values are similar.

4.4 The attenuation factor: correct ranking, overstated magnitude

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Table 3 compares the closed-form attenuation factor $A(t_{1/2})$, computed purely from each design's visit schedule, to the actual relative change in the fitted mixed-model coefficient's magnitude, $|\bar{\hat{\beta}}(t_{1/2}{=}1.0)| \,/\, |\bar{\hat{\beta}}(t_{1/2}{=}0)|$, from the same Monte Carlo cells. The direction is right in every case, and the ranking across designs is right: CO's off-drug visits are, on one randomization path, a pre-drug baseline period that carryover cannot contaminate at all, so CO shows the least closed-form attenuation of the three designs, exactly matching the compendium's own qualitative finding that CO's carryover-induced power loss is the smallest of the designs compared[1]. But the *magnitude* is badly overstated throughout: the closed form predicts a 24.5-percentage-point attenuation for Hybrid between $t_{1/2} = 0$

and $t_{1/2} = 1.0$ weeks, while the fitted coefficient's magnitude falls by only about 1.7% over the same range ($N = 70$: $|-0.2358|$ to $|-0.2317|$).

Table 3: Closed-form attenuation factor versus the actual relative change in the fitted interaction coefficient's magnitude, $N = 70$.

Design	$A(t_{1/2}{=}1.0)$ (closed form)	$\lvert\bar{\hat{\beta}}(1.0)/\bar{\hat{\beta}}(0)\rvert$ (simulated)
CO	0.973	1.031
Hybrid	0.755	0.983
OL+BDC	0.667	0.975

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The explanation is structural, not a fitting failure. The closed form's attenuation factor comes from a crude two-group mean split that discards each observation's individual timing; the fitted mixed model instead includes an explicit time covariate t alongside `bm:Dbc`, and recovers gradient information within the off-drug period (how the outcome trends as D_{bc} decays across successive off-drug visits) that the two-group contrast simply averages away. This is a genuine, useful negative result: it means the attenuation factor of Equation (3) should be read as a *design screening heuristic* — it will correctly tell an analyst which of two designs loses more information to a given carryover half-life — and not as a *quantitative bias estimate* for the coefficient a properly specified mixed-model analysis will actually report.

5 Design guidance for a mandatory open-label lead-in

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Given a fixed open-label first phase, the validated components above translate the compendium's qualitative guidance into a design procedure. First, compute $v_1(\text{design}, \rho)$ from Equation (5) for every design under consideration, using each candidate's actual visit schedule; this can be done for a design that has never been simulated, in seconds, without touching the simulation pipeline. Second, use v_1's ranking to eliminate designs that are clearly dominated (a large v_1 relative to the alternatives) and to identify a small shortlist worth a full Monte Carlo comparison; Table 1's validation shows this ranking is trustworthy for well-separated candidates and should not be trusted to resolve close calls, which still warrant simulation. Third, use $A(t_{1/2})$ from Equation (3) only as a relative screen across designs at a shared assumed half-life, not as a quantitative power adjustment; §4.4 shows it substantially overstates the actual attenuation a `corCAR1` mixed model with a time covariate will recover from. Fourth, since adding visits after the mandatory open-label phase is the largest lever in v_1's design-dependence (Table 1), and since the compendium's own matched-N comparison shows a brief crossover tail after blinded discontinuation is worth its cost in the one design pairing checked here[1], a Hybrid-style continuation of the mandatory open-label phase should be the default starting point for the design shortlist, to be displaced only by a full simulation showing a specific alternative outperforms it for the trial's actual parameters.

6 Limitations

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Several scope limitations bound how far this closed form should be trusted. It is derived and validated under the mean-moderation architecture only; the covariance-moderation architecture's interaction lives in $\partial \operatorname{Cor}(BM, BR)/\partial D_{bc}$ rather than in the conditional mean[1], and the two-group contrast reduction of Equation (1) does not apply to it without a separate derivation. The validation uses the Monte Carlo cells' own empirical $\overline{\hat{\beta}}$ as the effective true effect in Equation (7) rather than the architecture's nominal calibration target $c_{bm,A} \cdot \sigma_{BR}$: the two differ by roughly an order of magnitude in the data inspected here (3.6 nominal against approximately 0.23 observed), a discrepancy this communication does not resolve and flags as an open calibration question for the companion architecture manuscripts rather than papering over with an ad hoc rescaling. The variance formula's on/off independence approximation (Equation (5)) is shown, by its residual pattern in Table 1, to be imperfect for designs with long contiguous within-phase blocks. Validation is against a single ρ value (0.7, the driver default) and a single pair of sample sizes; whether the closed form's accuracy holds at other correlation strengths or substantially larger or smaller N is not established here. Finally, the closed form omits the placebo-belief and natural-history components of the outcome decomposition entirely[9]; a biomarker correlated with placebo responsiveness will bias the interaction estimate in a way no version of this closed form, restricted to the pharmacological channel alone, can detect.

7 Conclusion

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A closed-form approximation to the biomarker-treatment interaction estimator's sampling variance, built from a within-subject two-group contrast and the exact variance of a mean of AR(1)-correlated observations, reproduces 95% of the variation in the interaction estimator's Monte Carlo sampling variance and predicts simulated power to within a mean absolute error of 0.05, using a single fitted nuisance-variance parameter and no new simulation. A companion attenuation factor correctly ranks trial designs by their vulnerability to carryover but should not be read as a quantitative bias estimate, because it discards timing information a properly specified mixed-model analysis recovers. Together these give a concrete, quantitative starting point for the design question that motivated this communication — what to prioritize given a mandatory open-label first phase — without displacing the full Monte Carlo comparison the compendium's other manuscripts continue to supply for final design decisions.

8 Reproducibility

No new Monte Carlo simulation was run for this manuscript. All numbers reported here are produced by `analysis/scripts/nof1-design-efficiency/01-closed-form-validation.R`, which reads the existing output of `analysis/scripts/dgp-combined/01-run-combined-factorial.R` (`analysis/data/dgp-combined/01-combined-summary.rds`) and writes `analysis/data/nof1-design-efficiency/closed-form-validation.{rds,csv}`.

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